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# Trainer Management Module Documentation

## ## TrainerController – Overview

ASP.NET Core Controller for managing trainers with user integration, department assignment, and comprehensive reporting capabilities.

### Controller Configuration

| Setting       | Value                                     |
|---------------|-------------------------------------------|
| Area          | Admin                                     |
| Authorization | AdminAuthorize                            |
| Dependencies  | ITrainerService<br>IUnitOfWork<br>IMapper |

## ## Main CRUD Operations

### Index Action

| Aspect         | Description                                                  |
|----------------|--------------------------------------------------------------|
| Purpose        | Main trainer listing with user integration                   |
| Data Structure | Join between Trainer and User entities                       |
| Search         | Arabic/English trainer name, user-based filtering            |
| Includes       | Related Courses per trainer                                  |
| UI Features    | Pagination (50/page), AJAX partial views, department context |

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### AddEdit (GET)

| Feature               | Details                                              |
|-----------------------|------------------------------------------------------|
| Purpose               | Create or edit trainer form                          |
| User Selection        | Dropdown excludes users already assigned as trainers |
| Department Assignment | Department dropdown population                       |
| Mode                  | Auto-detect create vs edit using ID                  |

### AddEdit (POST)

| Aspect        | Description                                         |
|---------------|-----------------------------------------------------|
| Validation    | ModelState validation with dropdown state retention |
| File Handling | PDF validation, max size 5MB                        |
| Redirect      | Create → Index, Edit → same page                    |

### Delete

| Aspect | Description                             |
|--------|-----------------------------------------|
| Method | HTTP POST                               |
| Flow   | Service deletion then redirect to Index |

## ## Reporting & Export Features

### Print

| Property  | Description                             |
|-----------|-----------------------------------------|
| Purpose   | Printer-friendly trainer list           |
| Attribute | [IgnoreAction]                          |
| Behavior  | Same search as Index without pagination |

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Excel Export

| Column       | Description                |
|--------------|----------------------------|
| Trainer Name | Localized Arabic / English |
| Phone        | Trainer contact number     |
| Email        | User email / username      |
| Department   | Localized specialization   |

## TrainerService – Core Operations

| Method                        | Purpose                            |
|-------------------------------|------------------------------------|
| GetAllAsync()                 | Retrieve all trainers              |
| GetByIdAsync(int)             | Fetch trainer by ID                |
| GetByIdAsync_byuserId(string) | Resolve trainer by Identity UserId |

## Trainer Entity

| Property       | Type   | Description              |
|----------------|--------|--------------------------|
| Id             | int    | Primary key              |
| DepartmentId   | int    | Department foreign key   |
| UserId         | string | Linked Identity user     |
| Bio            | string | Trainer biography        |
| AttachmentPath | string | Certificates / documents |

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## Trainer DTO – TrainersNameDTO

| Property    | Description        |
|-------------|--------------------|
| Id          | Trainer identifier |
| UserId      | Associated user    |
| FullNameAr  | Arabic name        |
| FullNameEn  | English name       |
| Email       | Trainer email      |
| PhoneNumber | Contact phone      |

## Trainer ViewModels

| ViewModel         | Purpose                                         |
|-------------------|-------------------------------------------------|
| TrainerIndexVM    | Main listing with pagination and departments    |
| TrainerWithUserVM | Combines Trainer and User entities              |
| TrainerVM         | Create/Edit trainer with validation and uploads |
| TrainersNameVM    | Lightweight selection model                     |

## Key System Benefits

- Strong Identity integration
- Department-based specialization
- Secure permission-driven UI
- Print and Excel reporting
- Scalable trainer–course relationships